INCH-POUND
MIL-DTL-32525A
20 May 2022
SUPERSEDING
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19 January 2016

DETAIL SPECIFICATION

MAGAZINE, PORTABLE (SHIPBOARD)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable shipboard magazines for installation aboard Navy and commercial vessels. The portable magazine is intended for the safe and secure stowage of limited types and quantities of ammunitions and explosives. See 6.1 for intended use.

1.2 Classification. Portable magazines are of the following types and classes as specified (see 6.2).

a. Type I – For use on shock hardened combat vessels.

(1) Class A – Magazine to stow mission critical ordnance; grade A shock.

(2) Class B – Magazine to stow cargo ordnance, small arms ammunition, pyrotechnics, and miscellaneous ordnance; grade B shock.

(3) Class C – Airborne expendable countermeasures (AECM) assembly area; grade B shock.

b. Type II – For use on shock hardened non-combat vessels.

(1) Class A – Magazine to stow mission critical ordnance; grade A shock.

(2) Class B – Magazine to stow cargo ordnance, small arms ammunition, pyrotechnics, and miscellaneous ordnance; grade B shock.

(3) Class C – AECM assembly area; grade B shock.

c. Type III – For use on non-shock hardened vessels.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FEDERAL SPECIFICATIONS

FF-S-86 - Screw, Cap, Socket-Head

FF-S-92 - Screw, Machine, Slotted, Cross-Recessed or Hexagon Head

FF-S-200 - Setscrews: Hexagon Socket and Spline Socket, Headless

TT-P-645 - Paint, Alkyd Type, Formula Number 84

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

MIL-DTL-3316 - Adhesives, Fire-Resistant – for Thermal Insulation

MIL-PRF-8805/99 - Switches, Push, 2 Circuit and DPDT, Alternate and Momentary Action, 2 Amperes, Splashproof

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-DTL-15514 - Telephone Equipment, Sound Powered Telephone Handset, Headset-Chest Set, and Headset-Chest Set, Noise Attenuating

MIL-S-16032 - Switches and Detectors, Shipboard Alarm Systems

MIL-DTL-16377 - Fixtures, Lighting; and Associated Parts; Shipboard Use, General Specification for

MIL-DTL-16377/12 - Fixtures, Lighting; Fluorescent and SSL Equivalent, General Lighting, 60 Watts, 120 Volts, 60 Hertz, Watertight, Symbols 333.1, 333.2, 341.1, 341.2, 342.2, 342.3, 333.1SSL, 333.2SSL, 341.1SSL, 341.2SSL, 342.2SSL, and 342.3SSL

MIL-DTL-16377/27 - Fixtures, Lighting; Incandescent and SSL Equivalent, General Lighting 50 Watts, 120 Volts, Watertight, Symbols 92.2, 93.2, 93.4, 93.5, 92.2SSL, 93.2SSL, 93.4SSL, and 93.5SSL

MIL-DTL-16377/42 - Fixtures, Lighting; Filters, Plastic, Tubular Shaped for Fluorescent Lamps

MIL-DTL-16377/53 - Fixtures, Lighting; Incandescent and Light Emitting Diode (LED), Detail Lighting, Lantern, Hand, Portable and Bulkhead Mounted, Watertight, Symbols 100.2, 100.2L, 100.3, 100.3L, 101.2, 101.2L, 101.3, 101.3L, 102.2, 102.2L, 108, and 108L

MIL-F-16377/54 - Fixture, Bracket, Lug Supporting for Hand Lanterns Symbol 2438

MIL-DTL-16377/73 - Fixtures, Lighting; Fluorescent and SSL Equivalent, Emergency Lighting, 20 Watts, 120 Volts, 60 Hertz, Watertight, Symbols 331.1EM, 331.2EM, 347.2EM, 347.3EM, 331.1EMSSL, 331.2EMSSL, 347.2EMSSL, and 347.3EMSSL

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MIL-DTL-16377/74	-	Fixtures, Lighting; Fluorescent and SSL Equivalent, Emergency Lighting, 40 Watts, 120 Volts, 60 Hertz, Watertight, Symbols 77.4EM, 77.5EM, 344.2EM, 344.3EM, 77.4EMSSL, 77.5EMSSL, 344.2EMSSL, and 344.3EMSSL
MIL-DTL-16377/75	-	Fixtures, Lighting; Fluorescent and SSL Equivalent, Emergency Lighting, 60 Watts, 120 Volts, 60 Hertz, Watertight, Symbols 333.1EM, 333.2EM, 341.1EM, 341.2EM, 342.2EM, 342.3.EM, 333.1EMSSL, 333.2EMSSL, 341.1EMSSL, 341.2EMSSL, 342.2EMSSL, and 342.3.EMSSL
MIL-A-17196	-	Alarm Switchboards, Alarm Panels, Remote Sentry Alarm Switch, Signal Units, and Special Alarm Connection Box, Naval Switchboard
MIL-I-17244	-	Indicators, Temperature, Direct-Reading, Bimetallic, (3 and 5 Inch Dial)
MIL-C-17588	-	Circuit Breakers (Automatic - ALB-1) and Switch, Toggle (Circuit Breaker, Non-Automatic - NLB-1) Air, Insulated Housing, 125 Volts and Below, A.C. and D.C., (Naval Shipboard Use)
MIL-C-20079	-	Cloth, Glass; Tape, Textile Glass; and Thread, Glass and Wire-Reinforced Glass
MIL-PRF-23236	-	Coating Systems for Ship Structures
MIL-T-24270	-	Thermowells for Thermometers and Electrical Temperature Sensors General Specification for
MIL-DTL-24558	-	Terminal Boxes, Connection, for Electrical and Electronic Systems, General Specification for
MIL-PRF-24635	-	Coating System, Weather-Resistant, Exterior Use
MIL-DTL-24643	-	Cables, Electric, Low Smoke Halogen-Free, for Shipboard Use, General Specification for
MIL-T-24649	-	Telephone, Hardware, Sound Powered, General Specification for
MIL-DTL-24653	-	Hasp, High Security, Shrouded, for Shipboard Doors and Hatches Using High Security Padlock, General Specification for
DOD-V-24657	-	Valve Actuator, Direct Coupled, Gear Driven Electrically Powered for Naval Shipboard Propulsion and Auxiliary Systems (Metric)
MIL-S-24660	-	Sprinkler Head, 180 Degrees Full Cone
MIL-PRF-24667	-	Coating System, Nonskid, for Roll or Spray Application
MIL-C-24707/5	-	Castings, Ductile Iron and Austenitic Ductile Iron
MIL-PRF-24758	-	Conduit Systems, Flexible, Weatherproof
MIL-PRF-24775	-	Air Conditioning Fan Coil Units, Horizontal and Vertical Types, Naval Shipboard
MIL-PRF-32161	-	Insulation, High Temperature Fire Protection, Thermal and Acoustic
MIL-DTL-32613	-	Controller, Auxiliary System, Naval Shipboard Use
MIL-PRF-81733	-	Sealing and Coating Compound, Corrosion Inhibitive

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-167-2	-	Mechanical Vibrations of Shipboard Equipment (Reciprocating Machinery and Propulsion System and Shafting) Types III, IV, and V
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-648	-	Specialized Shipping Containers
MIL-STD-777	-	Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships
MIL-STD-1310	-	Shipboard Bonding, Grounding, and other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-1472	-	Human Engineering
MIL-STD-1623	-	Fire Performance Requirements and Approved Specifications for Interior Finish Materials and Furnishings (Naval Shipboard Use)
MIL-STD-1689	-	Fabrication, Welding, and Inspection of Ships Structure
MIL-STD-3020	-	Fire Resistance of U.S. Naval Surface Ships
MIL-STD-3038	-	Test Methods for Ballistic Defeat Materials

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

401-1973965	-	Switch, Door, Type IC/DM-1-0
803-5001054	-	Hanger Pipe, Design, Installation and Arrangements
803-5184182	-	Insulation, Passive Fire Protection - Installation Details
803-5184187	-	Tiedown Assembly (HI-Shock) for Pallets, Cradles, Containers
803-5184207	-	Immersion Tank White Phosphorus Projectile
803-6397261	-	Doors, Watertight, Individually Dogged Arrangement
803-8436635	-	Universal Tiedown Deck Stowage System
804-860159	-	Storeroom Details
804-1360106	-	Locker Details
804-1385781	-	Hangers, Pipe, for Surface Ships, Temperature to 425 F

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804-1640412 - Equipment Requiring Operating and Safety Instructions

804-2425891 - Batten, Adjustable, Stowage System

804-4563098 - Bins, Storeroom Stowage Aids

804-4563099 - Storeroom Stowage Aids, J & K Racks, E & L Counters

804-4563100 - Storeroom Stowage Aids Type A and Type B Drawers, Shelf and Type S Bins

804-4563102 - Storeroom Stowage Aids C, M, N & T Racks

804-4563103 - Storeroom Stowage Aids Details

804-8436629 - Labels

810-1385791 - Fittings, Air Test, Sounding, Flooding & Draining

810-1385992 - Flanges, CU.NI., 250 PSI, WOG, Welding Neck

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) INSTRUCTIONS

NAVSEAINST C3401.1 - Nuclear Survivability Design Standards for Surface Ships of the U.S. Navy (Confidential)

(Copies of this document are available online at www.navsea.navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

0948-LP-012-5000 - Standard Navy Valves

NAVSEA OP 4 - Ammunition and Explosives Safety Afloat

NAVSEA OP 3565 Vol. 2 - Electromagnetic Radiation Hazards (Hazards to Ordnance)

S6430-AE-TED-010 - Piping Devices and Flexible Hose Assemblies

S9074-AQ-GIB-010/248 - Requirements for Welding and Brazing Procedure and Performance Qualification

S9086-CH-STM-010/074 - NSTM Chapter 074 – Volume 1, Welding and Allied Processes

S9086-CJ-STM-010/075 - NSTM Chapter 075, Fasteners

S9086-PA-STM-010/430 - NSTM Chapter 430, Interior Communication Installations

S9086-VH-STM-010/635 - NSTM Chapter 635, Thermal, Fire, and Acoustic Insulation

S9086-XG-STM-010/700 - NSTM Chapter 700, Shipboard Arms, Ammunition, and Explosives Handling and Stowage

S9522-AA-HBK-010 - Magazine Sprinkling Systems

SW024-AA-IDS-010 - Portable Magazine Ship Interface Considerations

T9070-AJ-DPC-120/3010 - Shock Design Criteria for Surface Ships

T9070-BA-DPC-020/511-2 - Heat Transfer Coefficients

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T9070-BD-DPC-010/522-1 Shipboard Sprinkler Systems

T9500-AA-PRO-130/510 - Air Conditioning, Ventilation, and Heating of Surface Ships

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

OPNAV INSTRUCTIONS

OPNAVINST 5530.13 - United States Navy Conventional Arms, Ammunition and Explosives Physical Security Policy Manual

(Copies of this document are available online at https://www.secnav.navy.mil/doni/default.aspx.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.6.2 - Square Head Set Screws and Slotted Headless Set Screws (Inch Series)

(Copies of this document are available online at www.asme.org.)

ASSOCIATION FOR MATERIALS PROTECTION AND PERFORMANCE (AMPP)

SSPC Painting Manual, Vol. 2 - Systems and Specifications

SSPC-SP 2 - Hand Tool Cleaning

SSPC-SP 10/NACE No. 2 - Near-White Metal Wet Abrasive Blast Cleaning

SSPC-SP 11 - Power Tool Cleaning to Bare Metal

(Copies of these documents are available online at www.ampp.org.)

ASTM INTERNATIONAL

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM D4417 - Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel

ASTM F467 - Standard Specification for Nonferrous Nuts for General Use

ASTM F468 - Standard Specification for Nonferrous Bolts, Hex Cap Screws, Socket Head Cap Screws, and Studs for General Use

ASTM F593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs

ASTM F594 - Standard Specification for Stainless Steel Nuts

ASTM F718 - Standard Specification for Shipbuilders and Marine Paints and Coatings Product/Procedure Data Sheet

ASTM F837 - Standard Specification for Stainless Steel Socket Head Cap Screws

ASTM F1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment, and Facilities

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(Copies of these documents are available online at www.astm.org.)

BRITISH STANDARDS INSTITUTION (BSI)

EN 54-2 - Fire Detection and Fire Alarm Systems — Part 2: Control and Indicating Equipment

EN 54-5 - Fire Detection and Fire Alarm Systems, Part 5: Heat Detectors - Point Heat Detectors

EN 54-7 - Fire Detection and Fire Alarm Systems, Part 7: Smoke Detectors – Point Smoke Detectors that Operate Using Scattered Light, Transmitted Light or Ionization

(Copies of these documents are available online at www.bsigroup.com.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60529 - Degrees of Protection Provided by Enclosures (IP Code)

(Copies of this document are available online at www.iec.ch.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 72 - National Fire Alarm and Signaling Code

(Copies of this document are available online at www.nfpa.org.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors Used in Government Procurement

SAE AMS-STD-595/17925 - Insignia White

SAE AMS-STD-595/26270 - Medium Gray

(Copies of these documents are available online at www.sae.org.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 268 - Smoke Detectors for Fire Alarm Systems

UL 521 - Heat Detectors for Fire Protective Signaling Systems

UL 864 - Standard for Control Units and Accessories for Fire Alarm Systems

(Copies of these documents are available online at www.comm-2000.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3. When specified (see 6.2), a test report shall be provided.

3.2 Integration. The portable magazine shall be prefabricated for the intended use above the V-line including weather deck locations on Navy and non-Navy vessels. The portable magazine shall meet integration requirements with the host ship as specified by an Installation Control Drawing (ICD) (see SW024-AA-IDS-010).

3.3 Workmanship. External surfaces shall be free from burrs, slag, sharp edges, and corners except where sharp edges and corners are required. The internal stowage spaces shall be free from sharp protrusions that could damage the stowed ordnance or cause injury to personnel.

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3.4 Structure.

3.4.1 Physical dimensions. The portable magazine interior dimensions shall be sized to suit the quantities of ordnance to be stowed, as specified (see 6.2). When determining overall dimensions, all interior outfitting as specified herein, as well as the maximum exterior dimensions permitted for the intended location on the host ship, as specified (see 6.2), shall be considered. Minimum clear headroom shall be 6 feet, 5 inches in all areas within the magazine. Minimum allowance for fire insulation protected space shall be 3.5 inches. The size and quantities of portable magazines shall be determined based upon ordnance compatibility requirements, in accordance with NAVSEA OP 4, as specified (see 6.2). The weight of the portable magazine shall be determined by the size, as specified (see 6.2).

3.4.2 Design.

3.4.2.1 Materials. Portable magazine structure (excluding outfitting and stowage systems) shall be fabricated from steel. Perforated corrosion-resistant steel (CRES) sheathing and a watertight false deck capable of supporting all outfitting and stowages shall be installed, providing a protected space for fire insulation. Combustible material, including wood, shall not be used as part of the magazine structure. Interior finish materials and furnishings shall be in accordance with the fire performance requirements of MIL-STD-1623. All materials used shall be free from defects that would adversely affect the performance or maintainability of individual components or of the overall magazine assembly.

3.4.2.1.1 Extraneous outfitting. Unless otherwise specified (see 6.2), excluding structure (deck, bulkheads, overhead plating stiffeners, foundations, and load-bearing appendages, such as padeyes or tie-downs), all exterior components (including, but not limited to, sunshields, cowling, light shields, mounting brackets, fasteners, switch boxes, switchboards, equipment stowage, and mounting hardware) shall be CRES in accordance with ASTM A240/A240M, grades 316, 316L, or composites approved by the Technical Authority (see 6.4.7).

3.4.2.1.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4.2.1.2.1 Raw materials. Recovered materials, which have been collected or recovered from solid waste, may be reprocessed to become a source of raw materials, as opposed to virgin raw materials.

3.4.2.2 Loading. The portable magazine structure and foundation shall be designed to withstand the loads transmitted by environmental conditions, including dead loads (see 6.4.1), live loads (see 6.4.3 and 6.2), ship's motion (see 6.4.5 and 6.2), wave slap (see 6.4.8), wind loads (see 6.4.9), snow and ice loads (see 6.4.6), and any additional loadings, such as nuclear blast and thermal loads. Loads shall be transmitted into the host ship's structure, with no resulting deformation, separation, or other physical damage sustained by the portable magazine structure, foundation, or host ship structure, unless plastic deformation allowances are provided.

3.4.2.2.1 Unspecified loads. Structure for which loads are not explicitly specified shall be designed to withstand the loads that can reasonably be expected in service. Additional local stiffening, when specified (see 6.2), shall be installed to prevent excessive vibration, panting, or springing of plating.

3.4.2.2.2 Nuclear blast and thermal loads (type I only – see 1.2). When specified for U.S. Navy vessels (see 6.2), the magazine structure shall be designed to meet a specified nuclear blast and thermal loading using a specified ductility factor that will be provided by the Navy. The magazine structure shall be designed to transmit these loadings into the ship's structure and still remain attached to the ship. The nuclear effects protection level requirements shall be specified in the applicable ship specifications for the intended host ship. This requirement will depend on the protection level (Level 1, 2, or 3). The source document for nuclear blast and thermal engineering loadings is NAVSEAINST C3401.1. A label plate, in accordance with 3.4.2.22, shall be provided and installed indicating the protection level of the magazine.

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3.4.2.3 Shock. The shock requirements of the portable magazine, stowage equipment, and ship attachment points shall meet the classification requirements of 1.2, as specified (see 6.2), for deck mounted items in accordance with MIL-DTL-901 for shock hardened ships. A label plate in accordance with 3.4.2.22 shall be provided and installed indicating the shock requirement of the portable magazine. All sprinkling system components, high temperature alarms, and Fire Detection and Actuation System (FDAS) components (including Fire Alarm Control Units [FACU], FDAS smoke detectors, FDAS heat detectors, and associated modules) shall meet the requirements for grade A shock in accordance with MIL-DTL-901.

3.4.2.4 Vibration. The portable magazine subsystems and their installation methods, including all sprinkling components and FDAS components, shall be designed to withstand the expected dynamic vibration encountered in the service environment in accordance with MIL-STD-167-1.

3.4.2.5 Electromagnetic interference (EMI). All FDAS components shall meet the EMI compliance report requirements of MIL-STD-461, as specified (see 6.2), but the RS103 EMI test shall be conducted at 25 volts per meter instead of 10 volts per meter.

3.4.2.6 Allowable stresses. Under the design loads, stresses in structural members, excluding bending stresses in individual plating panels, shall not exceed the limits specified in 3.4.2.6.1 and 3.4.2.6.2.

3.4.2.6.1 Tensile and bending stresses. Where there is no danger of failure from instability, allowable limits for the algebraic sums of axial and bending stresses shall be as listed below:

a.	Steel, ordinary strength (OS):	28 kilopounds per square inch (ksi)
b.	Steel, higher strength (HS):	40 ksi
c.	Steel, high yield (HY-80):	55 ksi
d.	Steel, high yield (HY-100):	66 ksi
e.	Steel, high strength, low alloy (HSLA-65):	48 ksi
f.	Steel, high strength, low alloy (HSLA-100):	65 ksi

3.4.2.6.2 Shear stresses. Allowable limits for shear stresses shall be 60 percent of the allowable tensile stresses.

3.4.2.6.3 Combined bending and column stresses. The sum of the ratios of calculated stress to allowable stress shall not exceed unity.

3.4.2.7 Ship's deck interface. The magazine shall be designed to be either welded to the host ship's deck or mechanically secured with an attachment point system to the host ship's deck. The deck foundation/fittings (see 3.4.2.10), where applicable, shall be designed to the magazine loading in accordance with 3.4.2.2, 3.4.2.3, and 3.4.2.4, as applicable, and shall not exceed the specific deck loading capacity of the host ship. A label plate, in accordance with 3.4.2.22, shall be provided and installed indicating the maximum deck loading requirement for the portable magazine. The allowable deck loading of the host ship shall be as specified (see 6.2).

3.4.2.8 Ballistic and fragmentation protection. When specified (see 6.2), the portable magazine shall be configured so as to be readily fitted with additional armor to meet ballistic and fragmentation protection requirements. The armor protection may be designed to be integral with the magazine structural bulkhead and deck plating. The threat shall be as specified (see 6.2) by type and class in accordance with MIL-STD-3038 or other armor protection requirements, as applicable.

3.4.2.9 Lifting padeyes. Padeyes shall be provided for the lifting and handling of the portable magazine in the unloaded condition. Padeyes shall be designed with a safety factor of six, based on the ultimate strength of materials for safe working load. The direction of pull shall be made in the plane of the padeyes only.

3.4.2.10 Foundation/deck fittings. The base of the magazine shall be designed to support and secure the weight of the magazine and all contents stowed within. The base shall be designed as compact as practical to minimize the overall center of gravity of the magazine. The base shall accommodate forklift pockets (see 3.4.2.10.3) for portability of the magazine and shall be designed to interface with a deck-mounted foundation on the host vessel.

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3.4.2.10.1 Loading. All portable magazine and foundation deck fittings and structural fasteners shall be designed to withstand the loadings in accordance with 3.4.2.2 and shall be in accordance with the allowable stresses of 3.4.2.6.

3.4.2.10.2 Clearance. There shall be a clearance from the magazine exterior bottom to the top of the host ship's deck surface to allow for mechanical fastening or welding of the portable magazine foundation to the host ship's deck and for access to the portable magazine deck drain fittings.

3.4.2.10.3 Forklift pockets. The magazine shall be provided with forklift pockets. Forklift pockets shall be designed to support the weight of the portable magazine structure in the unloaded condition. Forklift pockets shall be designed in accordance with MIL-STD-648, except that the material shall be in accordance with ASTM A240/A240M, grade 304L or 316L.

3.4.2.11 Bulkheads and overhead. All portable magazine bulkheads, overhead, and stiffener structures shall be designed to withstand the loadings in accordance with 3.4.2.2, shall be in accordance with the allowable stresses of 3.4.2.6, and shall be not less than 14-gauge steel. Exterior structure shall be designed to avoid pockets where water from the environment can accumulate.

3.4.2.12 Door.

3.4.2.12.1 Door construction. Unless otherwise specified (see 6.2), the door shall be constructed in accordance with 803-6397261. The door shall be 26 by 66 inches.

3.4.2.12.2 Door swing. The door to the portable magazine shall open outward from the magazine. The door shall be readily accessible and operable in a seaway without unnecessary risk to the operator.

3.4.2.12.3 Hold-open device. A device designed to manually secure the door in the open position shall be provided to prevent closure when entering the magazine.

3.4.2.12.4 Grounding. The magazine door(s) shall be grounded. Grounding methods shall be in accordance with MIL-STD-1310. The grounding strap shall be installed from the adjacent magazine structure to the door and shall not interfere with the door's operation throughout the entire swing.

3.4.2.13 Steps. Removable steps for stair ladders shall be provided for access from the ship's deck up to the magazine deck for heights greater than 12 inches in accordance with MIL-STD-1472. The top step shall be level with the magazine deck and shall not interfere with the swing of the door. The steps shall be fabricated to be as lightweight as practical. The steps shall be designed to be secured to the magazine structure when in use and shall be capable of being removed and stowed inside the magazine when not in use.

3.4.2.14 Blow-out port. Portable magazines designed for the stowage of missiles, rockets, or other ordnance that produces high volumes of gas shall be designed to prevent overpressurization or equipped with relief ports to ensure that magazine internal air pressure does not reach dangerous levels. The location of the relief port shall be determined by the stowage location on the ship, as specified (see 6.2). The hatch or cover shall be fitted with hinges. CRES restraining cables shall also be provided to prevent the hatch or cover from becoming an airborne hazard in the event of hinge failure during actuation. The hatch or cover shall be provided with a release mechanism designed to release and allow the hatch or cover to open at an overpressure of one-half of the magazine's design pressure. The hatch cover mechanism shall be released by the overpressure from the inside only. The direction and orientation of the blow-out port shall be considered in the location of the portable magazine on the host ship.

The total blow-out clear area "A" shall be defined as A (sq ft) = 4.5*R/H where:

R is the burn rate in pounds per second (available from weapon program manager).

H is the minimum structural design pressure in static equivalent head in feet of seawater.

When required (see 6.2), blow-out relief port requirements shall be specified, including burn rate or weapon type. For magazines with no blow-out port provided, a label plate, in accordance with 3.4.2.22, shall be provided and installed indicating its prohibition for the stowage of missiles, rockets, or other ordnance that produces high volumes of gas.

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3.4.2.15 Welds/connections.

3.4.2.15.1 Welding. Welding of all magazine structures shall be of a continuous weld. Welding of extraneous outfitting, as defined in 3.4.2.1.1, exposed to the weather or any area that can hold water shall be of a continuous weld.

3.4.2.15.2 Welder certifications. The certification for welders and welding procedures shall be in accordance with the requirements of MIL-STD-1689 and S9074-AQ-GIB-010/248, as specified (see 6.2).

3.4.2.15.3 Fabrication, welding, and inspection. Fabrication, welding, and inspection of the portable magazine and its foundation on the host ship shall be in accordance with the requirements of MIL-STD-1689, as specified (see 6.2).

3.4.2.15.4 Mechanical fasteners. Threaded fasteners shall be installed in accordance with the guidance of S9086-CJ-STM-010/075. The following fastener materials are authorized: CRES grade 316, nickel-copper (such as Monel 400, or equal), and nickel-copper-aluminum (such as Monel K-500, or equal). The specifications listed in table I shall be used to procure the fasteners.

TABLE I. Fasteners.

Fastener Materials	Bolts, Studs, and Hex Capscrews	Nuts	Socket Head Capscrews	Machine Screws	Set Screws
CRES 316	ASTM F593 ASTM F837	ASTM F594	ASTM F594	FF-S-92	FF-S-200 ASME B18.6.2
Monel 400, or equal	ASTM F468	ASTM F467	FF-S-86	FF-S-92	FF-S-200 ASME B18.6.2
Monel K-500, or equal	MIL-DTL-1222	MIL-DTL-1222	MIL-DTL-1222	N/A	N/A

3.4.2.16 Fire performance. Portable magazines shall be considered as an N-30 boundary. N-30 boundary will provide fire protection against a rolling fuel fire on the deck that continues to burn under and around the portable magazine. All boundaries, including the deck, shall be classified as unrestricted N-30 with insulation on the unexposed side in accordance with MIL-PRF-32161 and MIL-STD-3020.

3.4.2.16.1 Fire insulation. Fire insulation shall be provided on the bulkheads, overhead, and between the false deck and the magazine bottom and shall be installed in general accordance with 803-5184182, or as approved by the Technical Authority.

3.4.2.16.2 Protection of fire insulation. Fire insulation on the bulkheads and overhead shall be located inside the magazine to protect against the weather. The insulation installed over the steel bulkheads and overheads shall be protected from damage due to impact from internal cargo and traffic with a perforated CRES sheathing.

3.4.2.16.3 Installation. All fire insulation shall be installed in general accordance with 803-5184182 or as approved by the Technical Authority and specific guidance provided in the drawings in fire test reports and approval letter associated with the selected N-30 fire resistant systems.

3.4.2.16.4 Door insulation. Fire insulation material shall be provided on the door to attain a fire rating of N-30. An A-60 door fire insulation rating may be acceptable for specific applications as approved by the Technical Authority.

3.4.2.17 Sunshields. Portable magazines shall be equipped with a removable sunshield on the roof to facilitate maintenance to the magazine top. When specified (see 6.2), additional sunshields may be provided for other surfaces. Sunshields shall meet all Radar Cross Section requirements where applicable, including vertical surface angles. Design details shall be in general accordance with 804-1360106. Other designs meeting the intent of 804-1360106 may be used as approved by the Technical Authority. CRES components shall be used for mounting fasteners.

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3.4.2.18 Temperature sensing. Portable magazine internal temperatures shall be monitored and recorded.

3.4.2.18.1 Bimetallic thermometer. A direct indicating bimetallic thermometer in accordance with the requirements of MIL-I-17244 shall be provided to monitor portable magazine internal temperature. The thermometer well shall be in accordance with the requirements of MIL-T-24270. The thermometer shall have a 3-inch dial indicating a temperature range of -140 to +180 °F, with resettable maximum and minimum temperature indicators. The thermometer shall be located to be read from outside of the magazine. 804-1360106 may be used as guidance. A corrosion-resistant watertight enclosure with a hinged door shall be mounted to the bulkhead adjacent to the access door to enclose the thermal well and cap.

3.4.2.18.2 Real time temperature sensor. When specified (see 6.2), a remotely monitored real time temperature sensor and digital display, as approved by the Technical Authority, may be provided in addition to the bimetallic thermometer. Digital displays shall be located with the FACU for the portable magazine. The system shall have a means to record minimum and maximum temperatures for each 24-hour period. Remote temperature sensing equipment shall meet the requirements of 3.9.

3.4.2.18.3 Temperature recording. A temperature record card holder shall be located inside the bimetallic thermometer watertight enclosure door (see 3.4.2.18.1) and adjacent to the real time temperature digital display (see 3.4.2.18.2), if provided.

3.4.2.19 Physical security.

3.4.2.19.1 High security hasps. Magazine doors shall be fitted with a high security hasp in accordance with MIL-DTL-24653.

3.4.2.19.2 Forced-entry resistance. The portable magazine shall be designed to resist forced entry (see 6.4.2) for a period of 5 minutes.

3.4.2.19.3 Openings/penetrations. Blow-out port or duct penetrations into the portable magazine greater than 96 square inches that may be subject to external tampering shall be fitted with an internal personnel obstruction grid made of ¾-inch diameter welded steel or aluminum bars with openings no larger than 4 inches in any dimension.

3.4.2.19.4 Door hinge pins/hinge security brackets. Magazine door(s) shall have nonremovable hinge pins. Where hinge pins are located external to the space, the door shall be fitted with one hinge security bracket per hinge to prevent opening the door if the hinge pins are removed. Brackets shall be installed in accordance with S9086-XG-STM-010/700. Brackets shall be installed on doors with standard dogs.

3.4.2.20 Compartment tightness. The portable magazine shall be designed and fabricated to be airtight to a design internal pressure of 2 pounds per square inch.

3.4.2.21 Preservation and painting.

3.4.2.21.1 Surface access. Preservation of steel structure and piping shall be accomplished prior to the installation of insulating materials.

3.4.2.21.2 Surface preparation.

3.4.2.21.2.1 Abrasive blasting. Prior to painting, blast clean all steel exterior and interior surfaces of the magazine bulkheads, structure, door, attachments, and sunshields thicker than 10-gauge steel in accordance with SSPC-SP 10/NACE No. 2.

3.4.2.21.2.2 Power tool preparation. Any steel thinner than 10-gauge, or steel that cannot be prepared in accordance with SSPC-SP 10/NACE No. 2, shall be prepared in accordance with SSPC-SP 11.

3.4.2.21.2.3 Hand tool preparation. Any non-ferrous piping, or any area that cannot be prepared in accordance with SSPC-SP 10/NACE No. 2, shall be hand tool prepared in accordance with SSPC-SP 2.

3.4.2.21.3 Coating systems. Coating systems shall be applied and cured in accordance with applicable Technical Authority-reviewed ASTM F718 sheets.

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3.4.2.21.3.1 Piping and machinery primer. All piping and machinery without commercial coating systems installed by the manufacturer shall be primed with a primer in accordance with TT-P-645 and shall be applied in two coats at between 1.5 and 3 mils dry film thickness per coat.

3.4.2.21.3.2 Interior coating system. Interior structure and components not exposed to weather shall be painted with a primer, stripe coat, and topcoat in accordance with MIL-PRF-23236, type V, VI, or VII, class 5 or 7. The primer shall be applied in one coat at 4 to 8 mils dry film thickness. The total final system shall be 8 to 16 mils dry film thickness. Final interior color for the magazine bulkhead, structure, insulation, and door shall be white in accordance with SAE AMS-STD-595, color number 17925.

3.4.2.21.3.3 Exterior coating system. Exterior structure and components exposed to weather shall be painted with a primer and stripe coat in accordance with MIL-PRF-23236, type V, VI, or VII, class 5 or 7. The primer shall be applied in one coat at 4 to 8 mils dry film thickness. A topcoat in accordance with MIL-PRF-24635, type V or VI, class 1 or 2 shall be applied in one coat at 5 to 8 mils dry film thickness. The total final system shall be 9 to 16 mils dry film thickness. Final exterior topcoat color for the portable magazine bulkhead, structure, and door shall be haze gray in accordance with SAE AMS-STD-595, color number 26270. Final exterior topcoat color for the sunshields shall be white in accordance with SAE AMS-STD-595, color number 17925.

3.4.2.21.4 Interior deck covering. Interior decks not exposed to weather shall have nonskid deck covering applied over the coating system in any area exposed to foot traffic in accordance with MIL-PRF-24667, type XI, composition PS.

3.4.2.21.5 Faying surfaces. All faying surfaces shall be preserved and exposed edges shall be sealed using a sealing compound to completely prevent water intrusion between the surfaces in accordance with MIL-PRF-81733, type I, class 2, grade B.

3.4.2.22 Label plates. Signs, placards, hazard markings, and label plates appropriate for the ordnance stowed and magazine operations in accordance with the appendix titled “Signs, Placards, Hazard Markings and Label Plates” in NAVSEA OP 4 shall be provided and installed, as specified (see 6.2). Plates shall be fabricated of CRES, or other non-metallic material. Plates shall be designed and secured in accordance with 804-8436629. The interface between the label plate and structure shall be considered a faying surface in accordance with 3.4.2.21.5.

3.4.2.22.1 Label plate holder. A label plate holder sized to accommodate 8½- by 11-inch placards shall be provided for ordnance stowage charts. These holders shall be constructed of CRES. The interface between the label plate holder and structure shall be considered a faying surface in accordance with 3.4.2.21.5.

3.4.2.22.2 Operating instructions charts. Operating instructions charts exposed to the weather shall be fabricated from anodized aluminum by the metal photo process in accordance with MIL-DTL-15024, type H. Where charts are installed on or proximate to topside surfaces with radar absorbent material, labels shall be of the pressure-sensitive type. The interface between the label plate holder and structure shall be considered a faying surface in accordance with 3.4.2.21.5.

3.5 Stowage systems. Portable magazines shall be fitted with restraint systems to prevent ordnance from coming adrift due to shock or ship movement under maximum sea state conditions.

3.5.1 Hand carry or handling equipment. Depending on the type of ordnance being stowed, the portable magazine shall be designed with a stowage system that is capable of stowing ordnance that is hand carried or brought in using material handling equipment (MHE), ordnance handling equipment (OHE), hoists, or other equipment approved to handle ordnances. For hand carried ordnance, stowage systems shall be designed to typically handle wooden and wire bound boxes and metal cans and containers of a size and weight within the limits of a two-person lift and carry in accordance with MIL-STD-1472 and ASTM F1166. If approved MHE or OHE is required to load the portable magazine, the deck, supporting structure, and the deck grid shall be designed to support the maximum concentrated wheel loading of required MHE or OHE with rated loads or the maximum weapon tiered load, whichever is greater. Concrete and a universal tiedown deck stowage system may be required for heavier ordnance requiring a higher deck load per square foot and shall be required for portable magazines that utilize MHE or OHE within the magazine.

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3.5.2 Cast iron. Semi-steel castings shall not be used in the construction of stowage equipment. Where iron castings are used, they shall only be of nodular graphitic iron in accordance with MIL-C-24707/5. Nodular graphitic iron castings shall be examined and tested in accordance with S9086-CH-STM-010/074.

3.5.3 Stowage system types. Selected stowage system(s), as specified (see 6.2), shall not interfere with sprinkler head spray coverage and may consist of the types specified in 3.5.3.1 through 3.5.3.4.

3.5.3.1 Adjustable batten. Portable batten stowage systems, consisting of deck grating, overhead grating strips, and spring-loaded adjustable telescoping battens, shall be in accordance with the applicable sections of 804-2425891. Adjustable battens securing the maximum ordnance load (see 6.4.4) shall be not stressed greater than 80 percent of their yield strength or deflect greater than $^{1}/_{240}$ times their length under maximum ship motion loads.

3.5.3.2 Bin and rack. Bin and rack stowage systems consisting of standard or modified bins for small item stowage shall be in accordance with the applicable sections of 804-860159 and 804-4563098 through 804-4563103. Bin shelves shall be perforated to allow water to drain through each shelf to the lower shelf. Each horizontal shelf surface shall have ¾-inch diameter holes spaced 2 inches on center in a diagonal pattern. Shelves shall be of sufficient strength to sustain a load of 500 pounds per unit section of shelf not exceeding 42 inches in width, or 200 pounds per square foot.

3.5.3.3 Universal tiedown deck stowage (UTDS) system. A UTDS system consisting of steel deck channels and latex concrete shall be in accordance with 803-8436635. The deck channels shall be placed on 9- or 12-inch centers. When used with the adjustable batten stowage system, the deck channel shall be placed on 9-inch centers. The UTDS system may be used with tiedown chains in accordance with 803-5184187 or the adjustable battens (see 3.5.3.1).

3.5.3.4 New designs. Stowage systems of new designs, including fixed stanchion systems, may be provided, subject to approval by the Technical Authority. New designs shall be fabricated to the minimum weight consistent with the service required. Reconfigurable equipment (such as battens) that will be handled manually shall be designed to facilitate installation and removal and shall be clearly marked for identification and usage.

3.5.4 Air space. The air space between the stowage stack and the surface of adjacent portable magazine bulkheads shall be not less than 2 inches.

3.5.5 Ordnance stowage arrangement. A blank ordnance stowage arrangement chart shall be provided showing the general arrangement of the space, as specified (see 6.2). A laminated drawing of the magazine space shall be posted on the magazine interior bulkhead using a holder (see 3.4.2.22.1).

3.5.6 White phosphorus stowage. If the magazine is intended to stow white phosphorus (WP) ordnance, as specified (see 6.2), an immersion soak tank shall be provided and sized to accept the ordnance type stowed. The tank shall be secured to the deck with quick release fasteners to allow for quick removal and disposal of contents. One portable WP ordnance immersion tank in accordance with 803-5184207 for immersing leaking WP munitions shall be installed in each WP magazine. The tank shall be watertight and provided with a lid that can be secured to the tank top.

3.5.7 AECM assembly. If the magazine is intended to be used as an AECM assembly area, the items specified in 3.5.7.1 through 3.5.7.5 shall be provided.

3.5.7.1 Assembly table. One or more electrically grounded assembly tables shall be provided for AECM assembly and disassembly. The dimensions of each table shall be not less than 24 inches by 24 inches. The table shall have a rim around the edges to prevent cartridges from rolling off during ship movement.

3.5.7.2 Ready-service stowage. Ready-service stowage shall be provided for built up flare buckets.

3.5.7.3 Heating, ventilation, and air conditioning (HVAC). An HVAC system appropriate for a manned space and the pyrotechnic devices used shall be provided. A dedicated exhaust system shall be provided.

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3.5.7.4 Jettison chute. If the portable magazine is intended for placement at the edge of the vessel, an overboard jettison chute shall be provided that is capable of jettisoning a full AECM dispenser module, shipping container, and a single AECM flare. If the portable magazine is not placed where a jettison chute can be used, a water immersion tank of similar construction to the tank in 3.5.6 shall be provided in the portable magazine and filled with water.

3.5.7.5 Sprinkling. The portable magazine shall be provided with an automated magazine sprinkling system in accordance with S9522-AA-HBK-010, except as modified below to incorporate actuation using the FDAS. One FDAS heat detector and not less than one sprinkler nozzle shall be located over the assembly table.

3.6 Handling systems. When specified (see 6.2), portable magazines shall be fitted with handling systems. Installed handling systems shall be in accordance with all applicable design requirements of NAVSEA OP 4.

3.7 Heating and cooling.

3.7.1 Heating and cooling loads. Determination of heating and cooling loads shall be in accordance with T9500-AA-PRO-130/510 or other means approved by the Technical Authority. For guidance in selection of heat transfer coefficients, see T9070-BA-DPC-020/511-2.

3.7.1.1 Magazine temperature. The portable magazine shall have a design operating temperature of 85 °F in the cooling season and 40 °F in the heating season.

3.7.1.2 Weather design conditions. The portable magazine shall be designed to operate in the following weather conditions: the cooling weather design conditions shall be a dry bulb temperature of 115 °F and a seawater temperature of 97 °F and the heating weather design conditions shall be a dry bulb temperature of -10 °F and a seawater temperature of 28 °F.

3.7.2 Equipment. Heating and cooling equipment shall be in accordance with 3.7.2.1 or 3.7.2.2, as specified (see 6.2). Electrical requirements for heating and cooling equipment shall be in accordance with 3.9.

3.7.2.1 Self-contained. Self-contained systems include the bulkhead units and split condenser/evaporator systems. The system shall be marine grade; refrigerant, if provided, shall be non-chlorofluorocarbon. Combination heating and cooling units may be provided.

3.7.2.2 Chilled water. Fan coil units with integral heaters in accordance with MIL-PRF-24775 may be provided.

3.7.2.2.1 Chilled water control valves. Chilled water systems, if provided, shall have control valves operated by thermostat.

3.7.2.2.2 Piping and material. Portable magazine chilled water cooling and drain system piping and components shall be in accordance with MIL-STD-777. Pipe supports shall be fabricated in accordance with 804-1385781.

3.7.2.2.3 Chilled water piping and equipment insulation. All chilled water piping, chilled water drain piping, and cold equipment shall be provided with anti-sweat insulation. Adhesives shall be compatible with the insulation. Lagging cloth shall be in accordance with MIL-C-20079. Adhesive shall be in accordance with MIL-DTL-3316. For interior piping, a coating system shall be applied to the lagging cloth in accordance with 3.4.2.21.3.2. For exterior piping, a coating system shall be applied to the lagging cloth in accordance with 3.4.2.21.3.3.

3.7.2.3 Thermostats. Thermostat(s) shall be provided to control both heating and cooling temperatures. Separate thermostats for heating and cooling may be provided. Thermostats internal to cooling/heating equipment may be provided.

3.7.2.4 Coil-fan interlock. All heaters and cooling coils shall be interlocked with their respective fans, if provided, so that they shall not be energized unless the fan is operating.

3.7.2.5 Condensate drain. Cooling system condensate shall drain through the deck or bulkhead via a check valve to the weather. Condensate shall not be allowed to collect in any portion of the structure.

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3.8 Fire protection requirements.

3.8.1 Sprinkling system, dry-type. A magazine sprinkling system designed to cool and protect ordnance stowed in the magazine shall be provided. The sprinkling system shall be the dry-type with a control valve (ball valve), electric actuator, and electronic FDAS. The system shall utilize seawater from the ship firemain for sprinkling. The dry-type magazine sprinkling system shall not contain seawater in piping downstream of the control valve when the system is in standby mode. All sprinkling system components shall be in accordance with the requirements of MIL-DTL-901, grade A; MIL-STD-167-1; and MIL-STD-461.

3.8.1.1 Sprinkling rates. Sprinkling rates shall be computed on an overhead basis allowing a minimum of 0.8 gallon per minute per square foot of magazine overhead area where the deck height is 8 feet or less. Where deck height exceeds 8 feet, an allowance of 0.1 gallon per minute per cubic feet of additional gross volume shall be made. For detailed instructions on sprinkler piping design, refer to T9070-BD-DPC-010/522-1. T9070-BD-DPC-010/522-1 shall be used for sprinkler system calculations.

3.8.1.2 Piping and material. Portable magazine fire protection system piping and components shall be in accordance with MIL-STD-777. Pipe supports shall be fabricated in accordance with 804-1385781.

3.8.1.3 Sprinkling system seawater supply connection. Seawater sprinkling shall be supplied from the magazine sprinkling control valve located inside the skin of the ship. The first connection shall be flanged between the magazine control valve and the temporary supply line (flexible hose, in accordance with S6430-AE-TED-010). The second connection shall be flanged between the temporary hose and the sprinkling system inlet port on the portable magazine. The second shall be installed no more than 12 inches from the bottom of the portable magazine. The flanged connections shall be in accordance with 810-1385992.

3.8.1.4 Sprinkler heads. Sprinkler heads in accordance with MIL-S-24660, or equivalent as approved by the Technical Authority, shall be installed. These heads shall be sized to provide the design density requirements calculated in accordance with MIL-S-24660. At the minimum spray radius, 1 foot below the nozzle, in accordance with MIL-S-24660, the design density shall be provided at 80 percent of the locations along the edge of the discharge ring. All sprinkler heads within a compartment shall be the same manufacturer, model, and orifice size.

3.8.1.5 Spacing and coverage. Sprinkler head coverage is dependent upon overlapping water spray onto the protected material from multiple directions and water application onto the bulkheads. Refer to S9522-AA-HBK-010 for general guidance for sprinkler head placement. S9522-AA-HBK-010 shall be used for specific guidance for providing full sprinkler coverage.

3.8.1.6 Clearances. Sprinkler heads shall be installed below fixed horizontal spray obstructions, such as pipes, vent ducts, beams, and lighting. Removable/portable partitions closer than 3 feet vertically below the sprinkler heads shall be considered as compartment boundaries. Sprinkler heads may be installed above the tracks for dunnage systems with portable stanchions, provided that the overhead tracks do not obstruct more than 30 percent of the horizontal cross-section of the sprinkler pattern (openings within the tracks may not be counted as open area). Sprinkler heads installed above overhead tracks shall be located between tracks and shall not be located within 4 inches vertically of the tracks to allow water distribution above the tracks.

3.8.1.7 Sprinkling drain. A portable magazine drain shall be provided through the deck within reach of an exterior bulkhead. The drain line shall have a check valve. The check valve shall be airtight to 4 ounces per square inch. The drain shall be routed through the deck to the weather. The drain shall be sized to handle the inlet flow from the sprinkling system. All material shall be in accordance with MIL-STD-777. Pipe supports shall be fabricated in accordance with 803-5001054.

3.8.1.8 Sprinkling control valve. An electrically operated magazine sprinkling control valve, ball type, shall be provided. The control ball valve shall be in accordance with 0948-LP-012-5000. The control valve shall be compatible with the control systems specified in 3.8.2. The valve actuator shall be in accordance with DOD-V-24657 with a maximum cycle time of 30 seconds. The valve and actuator shall have a manual override local to the valve that permits operation of the valve when power or control signals are lost. All sprinkling system components shall be in accordance with the requirements of MIL-DTL-901, grade A; MIL-STD-167-1; and MIL-STD-461.

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3.8.1.8.1 Flush connection. A flush capability shall be provided immediately downstream of the sprinkling control valve to permit flushing of the firemain through the control valve without flowing water through the sprinklers. The flush connection shall have a hose connection compatible with shipboard fire hoses. The flush connection shall be normally locked closed.

3.8.1.9 Water switch. A water switch (see 3.10.2) shall be installed immediately downstream of the sprinkling control valve flush connection to indicate water flow in the sprinkling piping. The alarm shall provide an audible and visual signal at the same location as the intrusion alarms.

3.8.2 Control systems. The portable magazine shall utilize both manual and automatic controls for actuating the magazine sprinkling and manual controls for securing the sprinkling system after activation. The magazine sprinkling system power source and electrical components shall be in accordance with the requirements of 3.9. Cybersecurity for controls shall be in accordance with MIL-DTL-32613, as specified (see 6.2).

3.8.2.1 Manual control pushbutton stations. Manual pushbutton activation stations shall be provided to open and close the sprinkling control valve. Each pushbutton assembly shall be watertight and contain two momentary contact, Double-Pole, Single Throw (DPST) normally open buttons and a protective cover in accordance with MIL-DTL-2212. All pushbutton stations shall be provided with the capability to be tamper-sealed.

3.8.2.2 Manual control systems. Local and remote manual control systems for the magazine sprinkling system shall be powered from the same source.

3.8.2.2.1 Local control. Local manual pushbutton controls for actuating the magazine sprinkling shall be provided. Local controls shall be located on the exterior side of the magazine bulkhead and adjacent to the magazine access door. The access to the space shall be visible from the local pushbutton activation station. If in the weather, the pushbutton shall be installed in a non-locking, quick-opening, weather-proof enclosure with a desiccant installed.

3.8.2.2.2 Remote control. Remote manual pushbutton controls for actuating the magazine sprinkling shall be provided for installation inside the skin of the ship on the same deck as the magazine and visible from the sprinkling control valve. For spaces below the damage control deck, a remote manual pushbutton activation station shall be installed on the Damage Control Deck.

3.8.2.2.3 Instruction plates. Instruction plates shall be posted by each manual control station. The instruction plate shall describe the operation of the manual controls. The interface between the instruction plate and structure shall be considered a faying surface in accordance with 3.4.2.21.5.

3.8.2.3 FDAS. The magazine sprinkling system shall have a self-contained FDAS, as approved by the Technical Authority, for detecting smoke and high temperatures and activating sprinkling based on high temperature.

3.8.2.3.1 Materials. The materials shall meet the interface, operational, and performance requirements specified herein. Mercury and sealing compounds shall not be used in FDAS components.

3.8.2.3.2 Operating temperature range. FDAS components shall operate properly at temperatures ranging from 32 to 120 °F.

3.8.2.3.3 Configuration. The FDAS shall consist of an FACU and its associated FDAS heat detectors (which provide alarm indications as well as sprinkling activation) and FDAS photoelectric smoke detectors (which provide only alarm indications). The FDAS may incorporate high temperature alarms (105 °F heat detectors) as specified in 3.10.1, which provide only alarm indications.

3.8.2.3.4 Detectors. The FDAS heat detectors, the FDAS smoke detectors, and the high temperature alarms (105 °F heat detectors) shall be installed in accordance with appendix A. For ordnance stowage systems that obstruct smoke and heat flow to overhead detectors, heat detectors shall be required below the structure except where all the following conditions are met:

a. Openings in the overhead stowage system material are 0.25 inch (0.064 centimeter) or larger in the least dimension.

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b. Thickness of the overhead stowage system material does not exceed the least dimension of the openings.

c. Openings constitute at least 70 percent of the area of the overhead stowage material.

Regardless of mounting location, all detectors shall be accessible for performing testing and maintenance of the detector. The detector heads and base assemblies shall obtain power from, and be compatible with, the FACU. The base assembly of the detectors shall be a two-conductor device whose power supply terminals are the same as the signaling terminals. All detectors shall be in accordance with the requirements of MIL-DTL-901, grade A; MIL-STD-167-1; and MIL-STD-461, but the RS103 EMI test shall be conducted at 25 volts per meter instead of 10 volts per meter.

3.8.2.3.4.1 FDAS heat detector for sprinkling actuation. At least two spot-type FDAS heat detectors shall be provided in each portable magazine. All heat detectors shall be in accordance with either UL 521 or EN 54-5 and shall be compatible with the FACU (see 3.8.2.3.5). The detector head shall have an IP22 rating in accordance with IEC 60529. When the temperature at the detector head exceeds the fixed temperature alarm threshold, which shall be set to a temperature between 135 and 140 °F, an alarm shall be indicated at the FACU, which shall in turn activate sprinkling of the magazine. The detector head shall not provide alarms for any rate of temperature rise. A fusible alloy shall not be provided as the operating member of a heat detector. Heat detectors shall be restorable after alarm, but only after a manual reset at the FACU is executed and the temperature at the heat detector is below the alarm threshold (i.e., heat detector alarms shall latch at the FACU).

3.8.2.3.4.2 Photoelectric smoke detector. FDAS spot-type photoelectric (light scattering) smoke detectors shall be provided in accordance with either UL 268 or EN 54-7 and be compatible with the FACU (see 3.8.2.3.5). Smoke detectors shall latch at the FACU; smoke detectors shall be restorable after alarm, but only after a reset at the FACU is manually executed and the detector is free of smoke. Smoke detector alarms shall not activate magazine sprinkling.

3.8.2.3.4.3 Detector base assembly. The FDAS detector base assembly shall consist of a base and a junction box and shall accept attachment of all FDAS heat and smoke detector heads. The base and the junction box may be provided as one piece. The junction box shall be provided with mounting tabs or brackets and two watertight stuffing tubes/cable glands. The junction box shall provide a conductive path for the shields of shielded cables to connect to the ship's hull to provide shield termination.

3.8.2.3.5 FDAS FACU. A single FDAS FACU shall be provided for each portable magazine. The FACU shall be installed in a section of the host ship that is continuously manned when underway and in port. If the FACU cannot be installed in a continuously manned space, the FACU shall be designed to send trouble and alarm information to a location that is continuously manned and where the output relays are supervised. The FACU shall be approved in accordance with either UL 864 or EN 54-2. The FACU shall have a minimum IP22 rating in accordance with IEC 60529. The FACU shall provide power to and monitor/communicate with various FDAS detectors. Glass shall not be used for the cover of an FACU observation opening. Touch screen displays shall not be used. The FACU shall be in accordance with the requirements of MIL-DTL-901, grade A; MIL-STD-167-1; and MIL-STD-461, but the RS103 EMI test shall be conducted at 25 volts per meter instead of 10 volts per meter.

3.8.2.3.5.1 Individual addressability. The FDAS FACU shall communicate with and monitor individually addressed FDAS devices/detectors. Detector cabling shall be configured in a looped signaling line circuit arrangement in accordance with NFPA 72, class A.

3.8.2.3.5.2 Capacity. The FACU shall provide power to and monitor various FDAS devices/detectors. The total quantity of these devices will vary depending on the particular installation. The FACU shall be capable of powering and monitoring a total of at least 50 devices/detectors.

3.8.2.3.5.3 FDAS output relays. A minimum of four programmable relays shall be provided. The minimum rating of the relays shall be 120 volts alternating current (VAC), 2 amperes. At least one relay shall be capable of the following functions:

a. Relay 1: Activate a fire suppression system. This relay shall be used to actuate sprinkling for the portable magazine. Upon alarm, this normally open relay shall latch closed for 5 seconds when any 135 to 140 °F heat detector on the FACU detector cabling alarms. This relay shall then return to the normally open position.

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b. Relay 2: 135 to 140 °F heat detector alarm. This normally open relay shall close and remain closed when any 135 to 140 °F FDAS heat detector on the FACU detector cabling alarms. Additional alarms from 135 to 140 °F FDAS heat detectors associated with this FACU shall not change the closed state of the relay. This relay shall later open only after all 135 to 140 °F FDAS heat detectors associated with this FACU have been manually reset (provided that all 135 to 140 °F FDAS heat detectors have reset from their alarm state). This relay shall be used for remote indication of 135 to 140 °F FDAS heat detector alarms and shall not affect any other relays or result in the actuation of sprinkling for any portable magazines. The alarm shall provide an audible and visual signal in a section of the host ship that is continuously manned when underway and in port.

c. Relay 3: Smoke detector alarm. This normally open relay shall close and remain closed when any FDAS smoke detector on the FACU detector cabling alarms. Additional alarms from FDAS smoke detectors associated with this FACU shall not change the closed state of the relay. This relay shall later open only after all FDAS smoke detectors associated with this FACU have been manually reset (provided that all smoke detectors have reset from their alarm state). This relay shall not affect any other relays or result in the actuation sprinkling for any portable magazine. The alarm shall provide an audible and visual signal in a section of the host ship that is continuously manned when underway and in port.

d. Relay 4: FACU or associated detector trouble indication. This normally open relay shall close and remain closed when any trouble indication occurs on the FACU, including those troubles associated with any FDAS detector/device on the detector cabling. Additional trouble indications associated with the FACU and its devices/detectors shall not change the closed state of the relay. This relay shall later open only after all troubles associated with the FACU and its devices/detectors have cleared. This relay shall not affect any other relays or result in the actuation of sprinkling for any portable magazine. The alarm shall provide an audible and visual signal in a section of the host ship that is continuously manned when underway and in port.

3.8.2.3.5.4 FDAS back-up power source. The back-up power source shall be capable of operating the FDAS in a non-alarm condition for a minimum of 24 hours and then be capable of operating the output relays and loads for a period of 15 minutes. See 3.9.5 for upstream power requirements.

3.8.3 Label plates for sprinkling systems and FDAS. Label plates shall include operating, maintenance, safety, and piping/wiring/system diagrams, as applicable for each system, and shall be in accordance with 804-1640412. The label plates installed at each magazine sprinkler valve shall identify the compartment designation, the sprinkler head manufacturer, model, orifice size, and minimum residual sprinkler head design pressure for each space served.

3.9 Electrical requirements.

3.9.1 General requirements for electrical equipment. Electric-powered equipment shall be restricted to those devices and accessories required to maintain the safety and security of the portable magazine, including alarms, communications, lighting, sprinkling, heating, and cooling. Electric-powered equipment shall be in accordance with the requirements of MIL-DTL-917, as specified (see 6.2). All devices, fixtures, and enclosures shall be watertight in accordance with MIL-STD-108.

3.9.2 Grounding. Exposed metal surfaces of electric-powered equipment shall be at ground potential. Grounding methods shall be in accordance with the requirements of MIL-STD-1310.

3.9.3 Enclosures. Terminal boxes shall be in accordance with the requirements of MIL-DTL-24558. Equipment enclosures shall be in accordance with the requirements of MIL-DTL-2036.

3.9.4 Cables. Cables shall be interconnected within enclosures (see 3.9.3). Cables shall be of unbroken length and be in accordance with the requirements of MIL-DTL-24643. All cables mounted on the portable magazine interior and exterior shall be protected by CRES conduit using CRES mounting hardware. Weatherproof flexible conduit, if provided, shall be in accordance with the requirements of MIL-PRF-24758.

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3.9.5 Power input. Electric-powered equipment shall operate from a 440-VAC or 115-VAC, 60-hertz, type I power input from the host ship in accordance with MIL-STD-1399-300-1. The portable magazine shall be provided with not less than two 115-VAC, 20-ampere circuit breakers mounted in a watertight enclosure on the bulkhead exterior. The 115-VAC circuit breakers shall have dual-source, vital power including Automatic Bus Transfer for uninterrupted power supply. Circuit breakers for 115-VAC power shall be in accordance with MIL-C-17588, type NLB-1. If 440-VAC power is provided, there shall be a local disconnect mounted on a ship's bulkhead in close proximity to the portable magazine.

3.9.6 Secondary power supplies and control panels. For equipment requiring a semiconductor-based (electronic) power supply or control panel, the power supply/control panel shall be mounted in a watertight enclosure on the portable magazine bulkhead exterior. Electronic equipment and associated enclosures, cabling, and terminal boxes shall be in accordance with the requirements of MIL-STD-461 for electromagnetic emissions and susceptibility and be installed to the same requirements as the tested configuration.

3.9.7 Bonding. For portable magazines mechanically attached to the ship's structure, electrical bonding shall be in accordance with the requirements of MIL-STD-1310.

3.10 Alarms.

3.10.1 High temperature alarm system. A 105 °F high temperature alarm system (Circuit F) shall be installed. At least two high temperature alarm heat detectors, as approved by the Technical Authority, shall be installed in each portable magazine. Installed sensors shall activate audible and visual alarms on an alarm switchboard, as approved by the Technical Authority. This alarm switchboard shall be installed in a section of the host ship that is continuously manned when underway and in port. When the temperature at the detector head exceeds the fixed temperature alarm threshold of 105 °F, an alarm shall be indicated at the alarm panel. A fusible alloy shall not be provided as the operating member of this high temperature alarm heat detector. High temperature alarm heat detectors shall latch at the alarm panel; high temperature alarm heat detectors shall be restorable after alarm, but only after a manual reset at the alarm panel is executed and the temperature at the heat detector is below the alarm threshold. Sensors shall be installed in the overhead in accordance with appendix A and arranged so they are accessible for maintenance and not exposed to damage or adversely affected by proximity of HVAC components. All high temperature alarm heat system components shall be in accordance with the requirements of MIL-DTL-901, grade A; MIL-STD-167-1; and MIL-STD-461.

3.10.2 Sprinkling alarm system. A sprinkling alarm system (Circuit FH) shall be installed in accordance with S9522-AA-HBK-010 to provide an indication that the sprinkling system has been actuated or that there is leakage past the sprinkling control valve. A MIL-S-16032, type IC/W water switch shall be installed on the underside of a horizontal run of sprinkling piping, immediately downstream of the sprinkling control valve and the associated test isolation valve. Circuit FH shall activate audible and visual alarms on an alarm switchboard installed in a section of the host ship that is continuously manned when underway and in port.

3.10.3 Intrusion detection system (IDS). The portable magazine IDS (Circuit DA) shall be installed in accordance with OPNAVINST 5530.13. A door switch in accordance with the requirements of MIL-S-16032, type IC/DM-1-0 and 401-1973965 shall be installed on the access door and shall activate audible and visual alarms on an alarm switchboard installed in a section of the host ship that is continuously manned when underway and in port, and from which a response force can be dispatched. In the event of primary power loss, the IDS shall be capable of automatically transferring to an independent backup power source that is capable of powering the system for at least 4 hours. Line supervision or two independent means of alarm signal transmission to the monitoring switchboard shall be provided. Transmission lines shall be protected from tampering. Door switches shall be provided with uninstalled security wires and seals. Wires and seals shall be installed following shipboard installation, adjustment, and test of the switch.

3.10.4 Alarm switchboard. An alarm switchboard in accordance with MIL-A-17196, or equivalent, as approved by the Technical Authority, shall be provided for monitoring and alarming the abnormal conditions sensed by the detectors specified in 3.10.1, 3.10.2, and 3.10.3. A switchboard with an appropriate number of lines for each ship-set of portable magazines shall be provided or made available from the host ship. The switchboard shall be located aboard the host ship in a remote location with continuous manning while underway and in port.

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3.11 Communications. Communications with the alarm switchboard station as specified in 3.10.4, and other stations of that circuit on the host ship, shall be provided as specified (see 6.2).

3.11.1 Sound-powered telephone system. The portable magazine may have sound-powered voice communications equipment. The sound-powered telephone handset shall be in accordance with the requirements of MIL-DTL-15514. The sound-powered telephone handset holder, cabinet, jackbox, jack, and plug shall be in accordance with the requirements of MIL-T-24649. Sound-powered telephone equipment shall be mounted on the portable magazine bulkhead exterior, adjacent to break side of the access door. The handset, holder, and cabinet shall be approximately 60 inches above standing surface. The sound-powered telephone system shall have a means to alert the operator of an incoming call and to alert stations called.

3.11.2 Electric-powered communications. An electric-powered telephone or other ship interior voice communication system approved by the Technical Authority may be provided exterior to the magazine, as specified (see 6.2), and shall be in accordance with the requirements of S9086-PA-STM-010/430.

3.11.3 Radio frequency (RF) equipment. RF equipment, independent of the magazine design, may be used to meet the communications requirement herein but shall comply with Hazards of Electromagnetic Radiation to Ordnance (HERO) safety requirements (see 4.5.59). RF equipment shall not be installed within the magazine.

3.12 Lighting.

3.12.1 White illumination. The portable magazine shall have a permanent overhead lighting fixture(s) capable of providing not less than 7 footcandles (75.32 lux, 88 lumens) illumination and lighting sufficient to allow reading of labels on containers therein. Watertight fluorescent or solid state light emitting diode (LED) lighting fixtures, in accordance with the requirements of MIL-DTL-16377, shall be used. Technical Authority-approved globe type MIL-DTL-16377/27 SYM 92.2 fixtures, including Energy Focus, Inc. EFO-X001-92.2-02 LED, or equal, may be used. Incandescent lighting shall not be used. New Emergency Automatic Lighting Systems (NEALS) shall not be used. Remote source lighting shall not be used. Lights shall be energized via a bulkhead-mounted SYM 780.4 toggle switch mounted on the portable magazine bulkhead interior, adjacent to the break side of the door, approximately 48 inches above the deck surface of the magazine. A SYM 792 door switch shall be provided on the break side of the door jamb to prevent white lights from energizing unless operating under lighted ship conditions.

3.12.2 Red illumination. The portable magazine shall have a permanent overhead red lighting fixture(s) of the same type as the white lighting fixture meeting the requirements of 3.12.1. Red and white lighting may be generated by the same or separate light fixtures. Technical Authority-approved globe type MIL-DTL-16377/27 SYM 92.2 fixtures, including Energy Focus, Inc. EFO-X001-92.2-02 LED, or equal, may be used. MIL-DTL-16377/12 SYM 333.1 or 333.2 fluorescent light fixtures, if provided, shall use fluorescent lamps with red filter tube in accordance with the requirements of MIL-DTL-16377/42 on the center lamp. Red and white lighting shall be energized via the same bulkhead mounted toggle switch specified in 3.12.1. Red lighting shall operate under door open or closed conditions. Red lighting shall be sufficient to allow reading of the labels on containers in the magazine.

3.12.3 Emergency illumination. The portable magazine shall be outfitted with a minimum of one battery-operated portable battle lantern in accordance with the requirements of MIL-DTL-16377/53, MIL-DTL-16377/73, MIL-DTL-16377/74, or MIL-DTL-16377/75. The battle lantern shall be mounted to the portable magazine bulkhead interior adjacent to the break side of the door using a SYM 2438 mounting bracket in accordance with the requirements of MIL-F-16377/54.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

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4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified herein or as specified in referenced test documents.

4.3 First article inspection. First article inspection shall be performed on the portable magazine in accordance with table II when a first article sample is required (see 3.1 and 6.2).

4.4 Conformance inspection. Conformance inspection shall be performed at a frequency, as specified (see 6.2). One sample shall be subjected to verification in accordance with table II.

TABLE II. First article and conformance inspections.

Title	Requirements	Verification	First Article	Conformance
Workmanship	3.3	4.5.1	X	X
Physical Dimensions	3.4.1	4.5.2	X	X
Materials	3.4.2.1	4.5.3	X	X
Loading	3.4.2.2	4.5.4	X	
Shock	3.4.2.3	4.5.5	X	
Vibration	3.4.2.4	4.5.6	X	
Electromagnetic Interference (EMI)	3.4.2.5	4.5.7	X	
Allowable Stresses	3.4.2.6	4.5.8	X	
Ship's Deck Interface	3.4.2.7	4.5.9	X	
Ballistic/Fragment Protection	3.4.2.8	4.5.10	X	
Lifting Padeyes	3.4.2.9	4.5.11	X	
Foundation Loading	3.4.2.10.1	4.5.12	X	
Foundation Clearance	3.4.2.10.2	4.5.13	X	
Forklift Pockets	3.4.2.10.3	4.5.14	X	
Bulkheads and Overhead	3.4.2.11	4.5.15	X	
Standing Water	3.4.2.11	4.5.16	X	X
Door	3.4.2.12	4.5.17	X	X
Steps	3.4.2.13	4.5.18	X	X
Blow-out Port	3.4.2.14	4.5.19	X	X
Welds/Connections	3.4.2.15	4.5.20	X	X
Fire Performance	3.4.2.16	4.5.21	X	X
Sunshields	3.4.2.17	4.5.22	X	X
Temperature Sensing	3.4.2.18	4.5.23	X	
High Security Hasp	3.4.2.19.1	4.5.24	X	X
Forced Entry Resistance	3.4.2.19.2	4.5.25	X	
Openings/Penetrations	3.4.2.19.3	4.5.26	X	X
Door Hinge Pins/Hinge Security Brackets	3.4.2.19.4	4.5.24	X	X
Compartment Tightness	3.4.2.20	4.5.27	X	X
Preservation and Painting	3.4.2.21	4.5.28	X	X
Label Plates	3.4.2.22	4.5.29	X	X
Stowage System	3.5, 3.5.1 through 3.5.4	4.5.30	X	
Ordnance Stowage Arrangement	3.5.5	4.5.31	X	X

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TABLE II. First article and conformance inspections – Continued.

Title	Requirements	Verification	First Article	Conformance
White Phosphorus Stowage	3.5.6	4.5.32	X	X
Handling Systems	3.6	4.5.61	X	X
Heating and Cooling Loads	3.7.1	4.5.33	X	
Heating and Cooling Equipment	3.7.2	4.5.34	X	X
Sprinkling System Type	3.8.1	4.5.36	X	X
Sprinkling Rates	3.8.1.1	4.5.36	X	X
Piping and Material	3.8.1.2	4.5.37	X	X
Sprinkling System Seawater Supply Connection	3.8.1.3	4.5.38	X	X
Sprinkler Heads	3.8.1.4	4.5.39	X	X
Spacing and Coverage	3.8.1.5	4.5.40	X	X
Clearances	3.8.1.6	4.5.41	X	X
Sprinkling Drain	3.8.1.7	4.5.42	X	X
Sprinkling Control Valve	3.8.1.8	4.5.43	X	X
Flush Connection	3.8.1.8.1	4.5.44	X	X
Water Switch	3.8.1.9	4.5.45	X	X
Manual Control Pushbutton Stations	3.8.2.1	4.5.46	X	X
Manual Control System	3.8.2.2	4.5.47	X	X
FDAS	3.8.2.3, 3.8.2.3.1 through 3.8.2.3.3	4.5.48	X	X
FDAS Detectors	3.8.2.3.4	4.5.49	X	X
FDAS FACU	3.8.2.3.5	4.5.50	X	X
Electrical Equipment	3.9.1	4.5.51	X	X
Grounding	3.9.2	4.5.52	X	
Enclosures	3.9.3	4.5.53	X	X
Cables	3.9.4	4.5.54	X	
Power Input	3.9.5	4.5.55	X	
Secondary Power Supplies	3.9.6	4.5.56	X	
Bonding	3.9.7	4.5.57	X	
Alarms	3.10	4.5.58	X	
Communications	3.11	4.5.59	X	X
Lighting	3.12	4.5.60	X	

4.5 Design verification.

4.5.1 Workmanship. Construction workmanship shall be visually inspected to determine compliance with 3.3.

4.5.2 Physical dimensions. Calculations shall be performed and measurements taken to determine compliance with 3.4.1.

4.5.3 Materials. Materials shall be visually inspected to determine compliance with 3.4.2.1.

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4.5.4 Loading. Finite Element Analysis (FEA), computer modeling and simulation, or calculations shall be conducted to determine compliance with 3.4.2.2.

4.5.5 Shock. The portable magazine with installed interior and exterior outfitting shall be tested in accordance with the requirements of MIL-DTL-901 or the Dynamic Design Analysis Method (DDAM) in accordance with T9070-AJ-DPC-120/3010 to determine compliance with 3.4.2.3 as required.

4.5.5.1 Installed handling systems shock. Installed handling systems shall be shock tested in the stowed position in accordance with the grade A requirements of MIL-DTL-901 or the DDAM in accordance with T9070-AJ-DPC-120/3010 to determine compliance with 3.4.2.3 as required.

4.5.6 Vibration. The portable magazine subsystems shall be vibration tested in accordance with MIL-STD-167-1 to determine compliance with 3.4.2.4.

4.5.6.1 Installed handling systems vibration. Installed handling systems shall be vibration tested in accordance with MIL-STD-167-1 or MIL-STD-167-2, as applicable, to determine compliance with 3.4.2.4.

4.5.7 EMI. The portable magazine subsystems, including all FDAS components and installed handling systems, shall be tested in accordance with MIL-STD-461 to determine compliance with 3.4.2.5.

4.5.8 Allowable stresses. FEA, computer modeling and simulation, or calculations shall be conducted to determine compliance with 3.4.2.6, 3.4.2.6.1, 3.4.2.6.2, and 3.4.2.6.3.

4.5.9 Ship's deck interface. FEA, computer modeling and simulation, or calculations shall be conducted to determine compliance with 3.4.2.7.

4.5.10 Ballistic and fragmentation protection. Test plates in representative thicknesses for added armor and magazine structure and door shall be tested in accordance with the requirements of MIL-STD-3038 for the threat, as specified (see 6.2), to determine compliance with 3.4.2.8.

4.5.11 Lifting padeyes. FEA, computer modeling and simulation, or calculations shall be conducted to determine compliance with 3.4.2.9.

4.5.12 Foundation loading. FEA, computer modeling and simulation, or calculations shall be conducted to determine compliance with 3.4.2.10.1.

4.5.13 Foundation clearance. Measurements shall be taken to determine compliance with 3.4.2.10.2.

4.5.14 Forklift pockets. Measurements shall be taken to determine compliance with 3.4.2.10.3.

4.5.15 Bulkheads and overhead. FEA, computer modeling and simulation, or calculations shall be conducted to determine compliance with 3.4.2.11.

4.5.16 Standing water. The magazine exterior shall be visually inspected to determine compliance with 3.4.2.11.

4.5.17 Door. The door shall be operated and visually inspected to determine compliance with 3.4.2.12.1, 3.4.2.12.2, 3.4.2.12.3, and 3.4.2.12.4.

4.5.18 Steps. The steps shall be fit tested and visually inspected to determine compliance with 3.4.2.13.

4.5.19 Blow-out port. FEA, computer modeling and simulation, or calculations shall be conducted to determine compliance with 3.4.2.14. The direction and orientation of the blow-out port shall be taken into consideration for the location on the host ship.

4.5.20 Welds/connections. Visual inspection shall be conducted to determine compliance with 3.4.2.15.1 and 3.4.2.15.4. Certification documentation shall be in accordance with 3.4.2.15.2 and 3.4.2.15.3, as specified (see 6.2).

4.5.21 Fire performance. Magazine interior vertical surfaces and overhead shall be visually inspected to determine compliance with 3.4.2.16, 3.4.2.16.1, 3.4.2.16.2, 3.4.2.16.3, and 3.4.2.16.4.

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4.5.22 Sunshields. Magazine sunshields shall be visually inspected to determine compliance with 3.4.2.17.

4.5.23 Temperature sensing. Operational testing and visual inspection shall be conducted to determine compliance with 3.4.2.18, 3.4.2.18.1, 3.4.2.18.2, and 3.4.2.18.3.

4.5.24 Physical security. The high security hasp on the portable magazine door shall be inspected and fit tested with the high security lock (provided by the ship) to determine compliance with 3.4.2.19.1. Visual inspection shall be conducted to determine that the door hinge pins are nonremovable type and that hinge security brackets are provided where hinge pins are located exterior to the space, in accordance with 3.4.2.19.4.

4.5.25 Forced entry resistance. A forced entry resistance (see 6.4.2) test shall be conducted to determine compliance with 3.4.2.19.2. The portable magazine shall resist forced entry for a period of 5 minutes through the door panel, hasp, hinges, or the weakest area of the magazine structure. The forcible entry test is conducted using battery-powered tools, manual tools, and heating/burning equipment. Battery-powered tools shall be limited to 18-volt capacity. Manual tools shall not exceed 18 inches in length or 3 pounds in weight. Heating/burning equipment shall be limited to single bottle systems. Total weight of all tools shall not exceed 35 pounds. Testing shall be conducted by one person.

4.5.26 Openings/penetrations. Visual inspection shall be conducted to determine compliance with 3.4.2.19.3.

4.5.27 Compartment tightness. Upon completion of all work in the portable magazine and before the painting of the magazine, a completion air pressure test shall be performed to demonstrate air tightness to determine compliance with 3.4.2.20. The portable magazine shall be pressurized to 4 ounces per square inch, maintained for 15 minutes for temperature stabilization, and shall then hold 4 ounces per square inch pressure for an additional period of 10 minutes. The portable magazine shall be considered sufficiently tight if the drop in air pressure does not exceed 2 ounces per square inch from the 4 ounces per square inch test pressure during the 10-minute test period. Air test fittings for the purpose of conducting this completion test shall be in accordance with 810-1385791 and shall be provided and located adjacent to the magazine door.

4.5.28 Preservation and painting. Visual inspection and measurement shall be conducted to determine compliance with 3.4.2.21 and all sub-paragraphs.

4.5.28.1 Surface profile. Surface profile shall be verified in accordance with method B or method C of ASTM D4417. One profile reading shall be taken for every 200 square feet. For method C, one profile reading shall be the average (mean) of three individual tapes.

4.5.28.2 Abrasive blasting. Prior to painting, near-white cleanliness shall be obtained with a surface profile between 2.0 and 4.0 mils with no reading exceeding 5.0 mils when tested in accordance with 4.5.28.1 to determine compliance with 3.4.2.21.2.1.

4.5.28.3 Power tool preparation. Prior to painting, power tool cleanliness shall be obtained with a surface profile not less than 2.0 mils when tested in accordance with 4.5.28.1 to determine compliance with 3.4.2.21.2.2.

4.5.28.4 Hand tool preparation. Prior to painting, hand tool cleanliness shall be obtained to determine compliance with 3.4.2.21.2.3.

4.5.28.5 Dry film thickness. Dry film thickness shall be verified in accordance with method PA 2 of the SSPC Painting Manual.

4.5.28.6 Piping and machinery primer. The dry film thickness per coat of primer shall be measured in accordance with 4.5.28.5 to determine compliance with 3.4.2.21.3.1.

4.5.28.7 Interior coating system. Primer and topcoat dry film thicknesses per coat shall be measured in accordance with 4.5.28.5. The total system dry film thickness shall be measured in accordance with 4.5.28.5. The final interior color for magazine bulkhead, structure, insulation, and door shall be tested to determine compliance with 3.4.2.21.3.2.

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4.5.28.8 Exterior coating system. Primer and topcoat dry film thicknesses per coat shall be measured in accordance with 4.5.28.5. The total system dry film thickness shall be measured in accordance with 4.5.28.5. The final exterior topcoat color for the portable magazine bulkhead, structure, and door shall be tested to determine compliance with 3.4.2.21.3.3. The final exterior topcoat color for the sunshields shall be tested to determine compliance with 3.4.2.21.3.3.

4.5.28.9 Nonskid deck covering. Visual inspection shall be conducted to determine compliance with 3.4.2.21.4.

4.5.28.10 Faying surfaces. All faying surfaces shall be verified by visual inspection for installation of a sealing compound in accordance with 3.4.2.21.5.

4.5.29 Label plates. Visual inspection shall be conducted to determine compliance with 3.4.2.22, 3.4.2.22.1, and 3.4.2.22.2.

4.5.30 Stowage systems. Visual inspection shall be conducted to determine compliance with 3.5.1, 3.5.2, 3.5.3, 3.5.3.1, and 3.5.3.2. FEA, computer modeling and simulation, or calculations shall be conducted to verify that the stowage systems are in compliance with 3.5.3.4. The stowage system will be inspected and tested for proper fit of anticipated ordnance loads and to verify compliance with cited requirements.

4.5.31 Ordnance stowage arrangement. Visual inspection shall be conducted to verify notional ordnance stowage arrangement drawing(s) are posted in accordance with 3.5.5, as specified (see 6.2).

4.5.32 White phosphorus stowage. Visual inspection shall be conducted to determine compliance with 3.5.6.

4.5.33 Heating and cooling loads. Calculations shall be conducted to determine compliance with 3.7.1, 3.7.1.1, and 3.7.1.2.

4.5.34 Equipment. Visual inspection and operational testing shall be conducted to determine compliance with 3.7.2, 3.7.2.1, 3.7.2.2, 3.7.2.3, 3.7.2.4, and 3.7.2.5.

4.5.35 Sprinkling system type. Visual inspection shall be conducted to determine compliance with 3.8.1.

4.5.36 Sprinkling rates. Calculations shall be conducted to determine compliance with 3.8.1.1.

4.5.37 Piping and material. Visual inspection shall be conducted to determine compliance with 3.8.1.2.

4.5.38 Sprinkling system seawater supply connection. Visual inspection shall be conducted to determine compliance with 3.8.1.3.

4.5.39 Sprinkler heads. Visual inspection shall be conducted to determine compliance with 3.8.1.4.

4.5.40 Spacing and coverage. Visual inspection shall be conducted to determine compliance with 3.8.1.5.

4.5.41 Clearances. Visual inspection shall be conducted to determine compliance with 3.8.1.6.

4.5.42 Sprinkling drain. Visual inspection shall be conducted to determine compliance with 3.8.1.7.

4.5.43 Sprinkling control valve. Visual inspection shall be conducted to determine compliance with 3.8.1.8. Verify by demonstration that the valve will cycle in less than 30 seconds and that the manual override permits operation of the valve in the absence of power.

4.5.44 Flush connection. Visual inspection shall be conducted to determine compliance with 3.8.1.8.1.

4.5.45 Water switch. Visual inspection shall be conducted to determine compliance with 3.8.1.9.

4.5.46 Manual control pushbutton stations. Visual inspection shall be conducted to determine compliance with 3.8.2.1.

4.5.47 Manual control systems. Local and remote manual controls shall be visually inspected and demonstrated to operate the control valve to determine compliance with 3.8.2.2, 3.8.2.2.1, and 3.8.2.2.2.

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4.5.48 FDAS. The FDAS shall be visually inspected to determine compliance with 3.8.2.3, 3.8.2.3.1, 3.8.2.3.2, and 3.8.2.3.3.

4.5.49 FDAS detectors. All heat and smoke detectors shall be inspected and subjected to sufficient heat or smoke to cause an alarm that shall register and latch at the FACU to determine compliance with 3.8.2.3.4, 3.8.2.3.4.1, 3.8.2.3.4.2, and 3.8.2.3.4.3.

4.5.50 FACU. The FACU shall be inspected and tested by simulating alarms, faults, and output relay conditions. The system shall be capable to alter configuration to permit complete testing (alarms, detectors, and ball valve) while in service without water flowing to sprinkler heads. System inspection and tests shall be conducted to determine compliance with 3.8.2.3.5, 3.8.2.3.5.1, 3.8.2.3.5.2, 3.8.2.3.5.3, and 3.8.2.3.5.4.

4.5.51 Electrical equipment. Visual inspection shall be conducted to determine compliance with 3.9.1. Equipment not previously certified for compliance with watertight requirements shall be tested in accordance with MIL-STD-108.

4.5.52 Grounding. Visual inspection shall be conducted to determine compliance with 3.9.2.

4.5.53 Enclosures. Visual inspection shall be conducted to determine compliance with 3.9.3.

4.5.54 Cables. Visual inspection shall be conducted to determine compliance with 3.9.4.

4.5.55 Power input. Visual inspection shall be conducted to determine compliance with 3.9.5.

4.5.56 Secondary power supplies and control panels. Inspection shall be conducted to determine compliance with the requirements of 3.9.6. Certification of compliance with MIL-STD-461 shall be provided, as specified (see 6.2).

4.5.57 Bonding. Visual inspection shall be conducted to determine compliance with 3.9.7.

4.5.58 Alarms. Inspection and testing shall be conducted to determine compliance with 3.10.1, 3.10.2, 3.10.3, and 3.10.4. Appropriate detector signatures including high temperature, water, flooded water, and intrusion shall be generated to ensure alarms are registered at the alarm panel. Appropriate faults shall be generated to ensure all are detected by the alarm panel.

4.5.59 Communications. Visual inspection and operational demonstration shall be conducted to determine compliance with the requirements of 3.11.1, 3.11.2, and 3.11.3. If RF equipment is planned for use, it shall comply with the safety requirements for HERO in accordance with NAVSEA OP 3565, Vol. 2.

4.5.60 Lighting. Visual inspection and operational demonstration shall be conducted and measurements taken as necessary to determine compliance with 3.12.1, 3.12.2, and 3.12.3.

4.5.61 Installed handling systems. Design calculations, FEA modeling, visual inspections, and testing shall be conducted to determine compliance with 3.6.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The portable magazine covered by this specification is intended to provide stowage for shipboard ammunition and explosives (A&E) in accordance with NAVSEA OP 4. The portable magazine is to be installed in an area of the host ship that optimizes its installation, ship impact, and usefulness. If more than one magazine is to be installed aboard the host ship, it is intended that they be co-located and separated for segregation purposes. The portable magazine is intended for ships with and without organic A&E stowage facilities. The portable magazine is not intended for the stowage of ammunition or explosives that require jettison capability or for returning ordnance that cannot be safely returned to its original packaging for stowage. The portable magazine is not intended for the stowage of ordnance containing lithium batteries, chemical A&E, hypergolic propellants, and other fuel-air explosive ammunition. The portable magazine is not intended for use as a shipping container. A&E should not be stowed in the portable magazine unless the magazine is installed on the host ship and all magazine support systems are connected and operational. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of portable magazine (see 1.2).
c. When first article inspection is required (see 3.1).
d. When first article test reports are required (see 3.1).
e. Quantities, sizes, individual weights, and compatibility requirements of all stowed material (see 3.4.1).
f. Portable magazine exterior limiting dimension(s) (see 3.4.1).
g. Total weight of stowed ordnance (see 3.4.2.2 and 6.4.3).
h. Ship’s motion factors (see 3.4.2.2 and 6.4.5).
i. Material type for exterior components (see 3.4.2.1.1).
j. Additional local stiffening requirements (see 3.4.2.2.1).
k. Nuclear blast and thermal load requirement (see 3.4.2.2.2).
l. Shock requirement of host ship (see 3.4.2.3).
m. EMI compliance reports for electronic powered equipment (see 3.4.2.5).
n. Allowable deck load of host ship’s deck (see 3.4.2.7).
o. Ballistic and fragmentation threat, if required (see 3.4.2.8).
p. Door design and construction if other than specified (see 3.4.2.12.1).
q. Blow-out relief port location (see 3.4.2.14).
r. Blow-out relief port requirements when required (see 3.4.2.14).
s. Verification of welder certification (see 3.4.2.15.2 and 4.5.20).
t. Fabrication, welding, and inspection requirements (see 3.4.2.15.3).
u. When additional sunshields for other surfaces than the roof may be provided (see 3.4.2.17).
v. When a real time temperature sensor is required (see 3.4.2.18.2).
w. Requirements for provision and installation of label plates for ordnance stowed (see 3.4.2.22).
x. Stowage system type desired (see 3.5.3).
y. Ordnance stowage arrangement chart (see 3.5.5).
z. When the magazine is intended to stow white phosphorus (see 3.5.6).
aa. When a handling system is required (see 3.6).
bb. Heating and cooling equipment requirements (see 3.7.2).

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cc. Cybersecurity of controls (see 3.8.2).

dd. MIL-DTL-917 requirements for electric-powered equipment (see 3.9.1).

ee. Communications equipment (see 3.11).

ff. When electric-powered telephone or other approved ship interior voice communication system is provided exterior to the magazine (see 3.11.2).

gg. Conformance inspection frequency (see 4.4).

hh. Testing of test plates when ballistic and fragmentation threat is specified (see 4.5.10).

ii. Requirement for posting of notional ordnance stowage arrangement drawing(s) (see 4.5.31).

jj. When certification of compliance with MIL-STD-461 is required (see 4.5.56).

kk. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be preproduction sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The contracting officer should also include specific Technical Authority approval of first article tests and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.1). Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

The requirement for first article inspection will be at the time of first contract for a discrete design within a specific classification of magazine (see 1.2) and when the contractor has not furnished the same discrete magazine design under this specification within a period of 48 months. The approval of the first article sample authorizes the commencement of shipment of the sample but does not relieve the supplier of responsibility for conformance acceptance.

6.4 Definitions.

6.4.1 Dead loads. Weight of all magazine structure, stowage fixtures, and support equipment.

6.4.2 Forced-entry resistance. The creation of a man-passable opening (96 square inches or greater as defined in MIL-HDBK-1013/1).

6.4.3 Live loads. Weight of all stowed ammunition and explosives and MHE/OHE, as specified (see 6.2).

6.4.4 Maximum ordnance load. 2,000 pounds distributed to 350 pounds per foot from deck grating to 73 inches above deck grating, per batten.

6.4.5 Ship's motion. The maximum dynamic forces produced by the motion of the ship in a seaway as specified in 6.2 or as defined by the more stringent of either the ship specification or DOD-STD-1399-301.

6.4.6 Snow and ice loads. The snow and ice load of 7.5 pounds per square foot as defined in MIL-STD-1399-302.

6.4.7 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.4.8 Wave slap. The load transmitted as a result of a wave slap of 500 pounds per square foot acting on the projected area of the portion of the magazine mounted on the weather deck and in accordance with the specific ship specifications and MIL-STD-1399-302.

6.4.9 Wind loads. The wind load of 30 pounds per square foot as defined in MIL-STD-1399-302.

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6.5 Part numbers. Contact information for the part numbers specified herein is as follows:

Energy Focus, Inc (CAGE 1BD15)
www.energyfocus.com
800-327-7877
32000 Aurora Road, Suite B
Solon, OH 44139

6.6 Subject term (key word) listing.

Ordnance

Security

Stowage

Weather

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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SMOKE AND HEAT DETECTOR INSTALLATION CRITERIA

A.1 SCOPE

A.1.1 Scope. This Appendix provides additional requirements for the placement of smoke and heat detectors within the portable magazine structure. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 SMOKE AND HEAT DETECTOR INSTALLATION

A.2.1 FDAS (135 °F) heat detector installation criteria. For portable magazines up to 10 feet in height, spacing criteria for heat detectors is as follows:

a. Detectors shall be mounted on the overhead of the compartment not less than 4 inches from the side of a beam, overhead stiffener, or bulkhead.

b. Detectors shall not be installed on the bottom of a beam or overhead stiffener.

c. The maximum area of coverage per heat detector shall not exceed 120 square feet.

d. The maximum spacing (S) between detectors shall not exceed 11 feet.

e. All points on the overhead shall have a detector within a distance equal to 0.7 times the maximum spacing (0.7S).

f. Detectors shall not be located within the airflow from an air supply diffuser or return air opening.

g. Detectors shall be located a minimum of 3 feet from an air supply diffuser, return air opening, or gravity cooling coil.

h. Overhead beam construction: Where beams or overhead stiffeners project more than 4 inches but less than 12 inches below the overhead, the spacing of heat detectors at right angles to the direction of beam or overhead stiffener travel shall be not more than two-thirds the maximum spacing. Where beams or overhead stiffeners project more than 12 inches below the overhead, each bay formed by the beams or overhead stiffeners shall be treated as a separate area requiring heat detectors. Where beams or overhead stiffeners project more than 4 inches below the overhead and the beams or overhead stiffeners are spaced less than or equal to 6.7 feet on center, the maximum spacing between detectors, measured parallel to the beam or overhead stiffener, shall not exceed 15 feet.

i. For compartments greater than 10 feet in height, heat detector linear spacing shall be reduced in accordance with table A-I prior to any additional reductions for overhead beams or overhead stiffeners.

TABLE A-I. FDAS (135 °F) heat and high temperature alarm system (105 °F) heat detector spacing reduction based on compartment height.

Overhead Height Above ___Feet	Up to and Including ___Feet	Up to and Including ___Feet
0	10	1.00
10	12	0.91
12	14	0.84
14	16	0.77
16	18	0.71
18	20	0.64
20	22	0.58
22	24	0.52
24	26	0.46
26	28	0.40
28	30	0.34

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A.2.2 FDAS smoke detector installation criteria.

a. Detectors shall be mounted on the overhead of the compartment not less than 4 inches from the side of an overhead beam, overhead stiffener, or bulkhead.

b. The maximum area of coverage per detector shall not exceed 500 square feet.

c. The maximum spacing (S) between detectors shall not exceed 22 feet.

d. All points on the overhead shall have a detector within a distance equal to 0.7 times the maximum spacing (0.7S).

e. Detectors shall not be located within the airflow from an air supply diffuser or return air opening.

f. Detectors shall be located a minimum of 3 feet from an air supply diffuser, return air opening, or gravity cooling coil.

g. Overhead beam construction: Where beams or overhead stiffeners project more than 4 inches but less than 12 inches below the overhead, the spacing of heat detectors at right angles to the direction of beam or overhead stiffener travel shall be not more than one-half the maximum spacing. Where beams or overhead stiffeners project more than 12 inches below the overhead, each bay formed by the beams or overhead stiffeners shall be treated as a separate area requiring smoke detectors. Smoke detectors may be mounted on the bottom of a beam or overhead stiffener that projects no more than 12 inches from the overhead.

A.2.3 High temperature alarm system (105 °F) heat detector installation criteria. For portable magazines up to 10 feet in height, spacing criteria for heat detectors is as follows:

a. Detectors shall be mounted on the overhead of the compartment not less than 4 inches from the side of an overhead beam, overhead stiffener, or bulkhead.

b. The maximum area of coverage per detector shall not exceed 250 square feet.

c. The maximum spacing (S) between detectors shall not exceed 15 feet.

d. All points on the overhead shall have a detector within a distance equal to 0.7 times the maximum spacing (0.7S).

e. Detectors shall not be located within the airflow from an air supply diffuser or return air opening.

f. Detectors shall be located a minimum of 3 feet from an air supply diffuser, return air opening, or gravity cooling coil.

g. Overhead beam construction: Where beams or overhead stiffeners project more than 4 inches but less than 18 inches below the overhead, the spacing of heat detectors at right angles to the direction of beam or overhead stiffener travel shall be not more than two-thirds the maximum spacing. Where beams or overhead stiffeners project more than 18 inches below the overhead, each bay formed by the beams or overhead stiffeners shall be treated as a separate area requiring heat detectors. Heat detectors may be mounted on the bottom of a beam or overhead stiffener that projects no more than 12 inches from the overhead.

h. For portable magazines greater than 10 feet in height, heat detector linear spacing shall be reduced in accordance with table A-I prior to any additional reductions for overhead beams or overhead stiffeners.

i. For each portable magazine, a minimum of two 105 °F heat detectors shall be installed, even if the portable magazine is 250 square feet or less.

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CONCLUDING MATERIAL

Custodians:
- Army – AV
- Navy – SH
- Air Force – 03
- DLA – CQ

Review activities:
- Army – GL, MI
- Navy – CG, OS
- Air Force – 69
- DLA – DH
- NGA – MP

Preparing activity:
- Navy – SH
- (Project 8145-2020-001)

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